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Chinese Remainder Theorem (Continued)

Q Solve $x \equiv 3 \pmod 7$, $x \equiv 5 \pmod{15}$

Soln Since $80 \equiv 3 \pmod 7$ and $80 \equiv 5 \pmod{15}$, so 80 is a solution.

$$\boxed{x \equiv 80 \pmod{105}}$$

$105 = 7 \times 15$
$\gcd(7, 15) = 1$

Method-I

5 mod 15 – Numbers congruent to 5 (mod 15)

5, 20, 35, 65, 80, 95 .....

mod 7 → 5, 6, 0, 1, 2, 3

So, $x \equiv 80 \equiv 3 \bmod 7$

$x \equiv 80 \equiv 5 \bmod 15$

Hence, $x \equiv 80 \pmod{105}$

Method-II (For large no.'s) → m, n

⇒ The numbers congruent to $b \pmod n$ are of the form $b + nk$ with k as integer. So we need to solve $b + nk \equiv a \pmod m$

$\Rightarrow nk \equiv (a-b) \pmod m$

Since $\gcd(m, n) = 1$. There is a multiplicative inverse i for $n \pmod m$. Multiplication by i gives

$\Rightarrow k \equiv (a-b)\, i \pmod m$

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Substituting back into $x \equiv b + nk$, then reducing mod m n gives the answer.

Ex - Solve $x \equiv 7 \pmod{12345}$, $x \equiv 3 \pmod{11111}$
(a, m, b, n)

Sol^n From previous example $11111x \equiv 4 \pmod{12345}$
The inverse of $11111 \pmod{12345}$ is $i = 2471$
Therefore,

$$k \equiv (a-b)i \pmod{m}$$
$$\equiv (7-3)\,2471 \pmod{12345}$$
$$\equiv 4 \cdot 2471 \pmod{12345}$$
$$\equiv 9884 \pmod{12345}$$

This yields $x \equiv (3 + 11111 \cdot 9884) \equiv 109821127 \pmod{11111 \cdot 12345}$

Chinese Remainder Theorem (General Form)

⇒ Let $m_1, m_2, \ldots m_k$ be integers with $gcd(m_i, m_j) = 1$ whenever $i \neq j$. Given integers $a_1, \ldots a_k$, there exists exactly one solution $x \pmod{m_1 \ldots m_k}$ to the simultaneous congruences

⇒ $x \equiv a_1 \pmod{m_1}$, $x \equiv a_2 \pmod{m_2}$ ....,
$x \equiv a_k \pmod{m_k}$

$$m = \prod_{i=1}^{k} m_i$$

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Ex- $m = m_1 . m_2 . m_3 = 105$ with $m_1 = 3$, $m_2 = 5$, $m_3 = 7$.
$a_1 = 2$, $a_2 = 2$, $a_3 = 3$

Soln The least multiplicative inverse of $m_1 = \frac{m}{m_1}$

$= 35 (\text{modulo } m_1 = 3)$

$= \underline{2} \rightarrow N_1$

The least multiplicative inverse of $m_2 = \frac{m}{m_2}$

$= 21 (\text{mod } m_2 = 5)$ is $\underline{1}$

$\downarrow$

N_2

The least multiplicative inverse of $m_3 = \frac{m}{m_3}$

$= 15 \text{ modulo } m_3 = 7 = \underline{1} \rightarrow N_3$

$$x = \sum_{j=1}^{3} a_j . N_j . m_j = 2.2.35 + 2.1.21 + 3.1.15 = \underline{227}$$

By reducing $x = 227$ mod $N = 105$, we get $\underline{x = 17}$.

For $x = 17$,

$x \equiv a_1 (\text{mod } m_1) \equiv a_2 (\text{mod } m_2) \equiv a_3 (\text{mod } m_3)$

$2 (\text{mod } 3) \equiv 2 (\text{mod } 5) \equiv 3 (\text{mod } 7) \equiv$

$227 (\text{mod } 105) \equiv 17 \text{ mod } 105$